



For the use of a Registered Medical Practitioner or a Hospital or a Laboratory only.

CISCORD INJECTION 1mg/ml

Cisplatin Injection 1 mg/ml

PRODUCT NAME

Cisplatin Injection 1 mg/ml

NAME AND STRENGTH OF ACTIVE INGREDIENT(S)

Cisplatin 1 mg

PRODUCT DESCRIPTION

Vials containing a clear, colourless to pale yellow solution.

After dilution: Clear colourless solution

PHARMACODYNAMICS / PHARMACOKINETICS

PHARMACODYNAMICS

Cisplatin has biochemical properties similar to those of bifunctional alkylating agents. The drug inhibits DNA synthesis by producing intrastrand and interstrand cross links in DNA. Protein and RNA synthesis are also inhibited to a lesser extent.

Although the principal mechanism of action of cisplatin appears to be inhibition of DNA synthesis, other mechanisms, including enhancement of tumour immunogenicity, may be involved in its antineoplastic activity. Cisplatin also has immunosuppressive, radiosensitising, and antimicrobial properties.

Cisplatin does not appear to be cell cycle specific.

PHARMACOKINETICS

There is good uptake of cisplatin by the kidneys, liver and intestine. More than 90% of platinum containing species remaining in the blood are bound (possibly irreversibly) to plasma proteins.

Penetration into the CSF is poor although significant amounts of cisplatin can be detected in intracerebral tumours.

The clearance of total platinum from plasma is rapid during the first four hours after intravenous administration, but then proceeds more slowly because of covalent binding to serum proteins. Levels of unbound platinum fall with a half-life of 20 minutes to 1 hour depending on the rate of drug infusion.

The elimination of intact drug and various platinum-containing biotransformation products is via the urine. About 15-25% of administered platinum is rapidly excreted in the first 2-4 hours after administration of cisplatin. This early excretion is mostly of intact cisplatin. In the first 24 hours after administration, 20-80% is excreted, the remainder representing drug bound to tissues or plasma protein.

INDICATION

Cisplatin injection is indicated for the palliative treatment of metastatic non-seminomatous germ cell carcinoma, advanced-stage and refractory ovarian carcinoma, advanced-stage and refractory bladder carcinoma and refractory squamous cell carcinoma of the head and neck.

It may be used as a single agent or combination with other chemotherapeutic agents.

It may be employed, in appropriate circumstances, in addition to other modalities, e.g. radiotherapy or surgery

RECOMMENDED DOSE & MODE OF ADMINISTRATION

ADULTS AND CHILDREN

Cisplatin should be administered by IV infusion over a 6-8 hour period. The recommended dose of cisplatin in adults and children is 50 to 100 mg/m² as a single IV dose every 3 to 4 weeks, or 15 to 20 mg/m² intravenously daily for 5 days every 3 to 4 weeks.

Interaction with aluminium:

Cisplatin may interact with metal aluminium to form a black precipitate of platinum. All aluminium-containing IV sets, needles, catheters and syringes should be avoided.

(1) Pretreatment Hydration:

Pretreatment hydration is required to induce diuresis during (and after) cisplatin administration. This hydration is achieved by giving 2 liters of either 0.9% sodium chloride or dextrose 4% in one-fifth normal saline (0.18%) over a 2-hour period.

During the last 30 minutes of the pre-treatment hydration or after the hydration, administer by side arm drip 37.5 g of mannitol (i.e., 375 ml of Mannitol 10% Injection).

(2) Preparation of Cisplatin Infusion:

Cisplatin 1 mg/ml Sterile Concentrate should be diluted in 2 litres of 0.9% sodium chloride injection. Do not refrigerate solutions.

(3) Treatment:

Following prehydration, administer the cisplatin infusion over 1 to 2 hours. It has been proposed that a longer infusion time of 6 to 8 hours may decrease the gastrointestinal and renal toxicities. The container should be covered to exclude light. Discard remaining contents after use.

(4) Post Treatment Hydration:

Continue IV hydration with the aim of administering another 2 litres of sodium chloride 0.9% injection, or dextrose 4% in sodium chloride 0.18% injection, over a period of 6 to 12 hours. Adequate hydration should be

maintained for 24 hours following infusion.

CONTRAINDICATION

Cisplatin may give allergic reactions in some patients. Use is contraindicated in those patients with a history of allergic reaction to cisplatin or other platinum containing compounds. Cisplatin induces nephrotoxicity which is cumulative. It is therefore contraindicated in patients with renal impairment.

Cisplatin has been shown to be cumulatively ototoxic and should not be given to patients with hearing impairment. Cisplatin is also contraindicated in myelosuppressed patients.

WARNINGS AND PRECAUTIONS

This agent should only be administered under the direction of oncologists in specialist units under conditions permitting adequate monitoring and surveillance. Supportive equipment should be available to control anaphylactic reactions.

Cisplatin reacts with metallic aluminium to form a black precipitate of platinum. All aluminium containing IV sets, needles, catheters and syringes should be avoided.

The solution for infusion should not be mixed with other drugs or additives.

Cisplatin produces cumulative nephrotoxicity which may be potentiated by aminoglycoside antibiotics. The serum creatinine, plasma urea or creatinine clearance and magnesium, sodium potassium, and calcium levels should be measured prior to initiating therapy, and prior to each subsequent course. Cisplatin should not be given more frequently than once every 3-4 weeks (see Undesirable Effects).

Anaphylactic-like reactions to cisplatin have been reported. These reactions have occurred within minutes of administration to patients with prior exposure to cisplatin and have been alleviated by administration of adrenaline, steroids and antihistamines.

Neurotoxicity appears to be cumulative. Prior to each course, the absence of symptoms of peripheral neuropathy should be established. Neurological examination should also be performed regularly (see Undesirable Effects).

Since ototoxicity of cisplatin is cumulative, audiometric testing should be performed prior to initiating therapy and prior to each subsequent course of the drug (see Undesirable Effects).

Peripheral blood counts should be monitored weekly. Liver function should be monitored periodically (see Undesirable Effects).

Adequate pre-treatment and 'during treatment' hydration should be ensured and such agents as mannitol given to minimise hazards of renal toxicity. In addition, adequate post-treatment hydration and urinary output should be monitored.

During, and for at least three months after therapy, both male and female patients should take contraceptive measures. As is the case with all anticancer drugs, in men this drug may cause transitional or permanent sterility. Preservation of sperm may be considered for the purpose of later fatherhood (see Pregnancy and lactation).

INTERACTIONS WITH OTHER MEDICAMENTS

Cisplatin can be used in combination with other cytostatics with corresponding mechanisms of action. The nephrotoxicity, ototoxicity and myelosuppression induced by cisplatin will be additive to existent impairment.

Nephrotoxicity might be exacerbated aminoglycoside antibiotics, administered simultaneously or 1-2 weeks after treatment with cisplatin (e.g. gentamicin).

Treatment with ototoxic drugs, like aminoglycoside antibiotics or loop-diuretics (e.g. furosemide), might increase the ototoxic potential of cisplatin, especially in impaired renal function. Ototoxicity may be enhanced with prior simultaneous cranial irradiation and may be related to peak plasma concentration of cisplatin. The literature indicates that cisplatin may reduce renal excretion of bleomycin and methotrexate (probably due to cisplatin-induced nephrotoxicity), and thus increase their toxicity.

Anti-epileptics: In patients receiving cisplatin and phenytoin, the serum level of phenytoin might be reduced. This is probably due to reduced absorption and/or increased metabolism. In these patients, one should monitor the levels of phenytoin in plasma, and adjust the dose accordingly.

Cisplatin may interact with aluminium. (See recommended dose & mode of administration)

PREGNANCY AND LACTATION

Use in Pregnancy:

There are no adequate data from the use of cisplatin in pregnant women. Cisplatin has been shown to be mutagenic in bacteria. It produces chromosome aberrations in tissue-cultures of animal cells and is teratogenic and embryotoxic in mice. Cisplatin should not be used during pregnancy unless the clinician considers the risk in an individual patient to be clinically justified.

Male and female patients should take contraceptive measures during, and for at least three months after therapy.

Use in Lactation:

Cisplatin should not normally be administered to mothers who are breastfeeding.

UNDESIRABLE EFFECTS

Nephrotoxicity: Renal toxicity has been noted in about one third of patients given a single dose of cisplatin when prior hydration has not been employed. It is first noted during the second week after a dose and is manifested by elevations in plasma urea and serum creatinine, serum uric acid and/or decrease

Cisplatin Injection 1mg-PIL (Mal)

Size : 140 x 210 mm (Front)

Colour : Pantone Black

Date : 02*05*16, 04*05*16 (1), 05*05*16

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	Packaging Dev.	Packaging Dev.	C.Q.A.	C.Q.A.
Signature				
Date				

in creatinine clearance.

Renal toxicity becomes more prolonged and severe with repeated courses of the drug. Renal function must return to acceptable levels before another dose of cisplatin can be given.

Renal function impairment has been associated with renal tubular damage. The administration of cisplatin using a 6-8 hour infusion with intravenous hydration and mannitol has been used to reduce nephrotoxicity. However renal toxicity still can occur after utilisation of these procedures.

Gastrointestinal toxicity: Nausea and vomiting occur in the majority of patients, usually starting within 1 hour of treatment and lasting up to 24 hours. Anorexia, nausea and occasional vomiting may persist for up to a week.

Ocular Toxicity: There have been reports of optic neuritis, papilloedema and cerebral blindness following treatment with cisplatin. Improvement and/or total recovery usually occurs following immediate discontinuation. Blurred vision and altered colour perception have been reported after the use of regimens with higher doses of cisplatin or greater dose frequencies than those recommended.

Ototoxicity: Ototoxicity has occurred in up to 31% of patients treated with a single dose of cisplatin 50 mg/m². Ototoxicity may be more severe in children and more frequent and severe with repeated doses.

Careful monitoring should be performed prior to initiation of therapy and prior to subsequent doses of cisplatin.

Unilateral or bilateral tinnitus, which is usually reversible, and/or hearing loss in the high frequency range, may occur.

The overall incidence of audiogram abnormalities is 24%, but large variations exist. These abnormalities usually appear within 4 days after drug administration and consist of at least a 15 decibel loss in pure tone threshold. The damage seems to be cumulative and is not reversible. The audiogram abnormalities are most common in the 4000-8000 Hz frequencies.

Haemotoxicity: Myelosuppression is observed in about 30% of patients treated with cisplatin. Leucopenia and thrombocytopenia are more pronounced at higher doses. The nadirs in circulating platelets and leucocytes generally occur between days 18-23 (range 7.3 to 45) with most patients recovering by day 39 (range 13 to 62). Leucopenia and thrombocytopenia are more pronounced at doses greater than 50 mg/m². Anaemia (decreases of greater than 2 g% haemoglobin) occurs at approximately the same frequency, but generally with a later onset than leucopenia and thrombocytopenia. Subsequent courses of cisplatin should not be instituted until platelets are present at levels greater than 100,000/mm³ and white cells greater than 4,000/mm³. A high incidence of severe anaemia requiring transfusion of packed red cells has been observed in patients receiving combination chemotherapy including cisplatin.

Anaphylaxis: Reactions possibly secondary to cisplatin therapy have been occasionally reported in patients who were previously exposed to cisplatin. Patients who are particularly at risk are those with a prior history or family history of atopy. Facial oedema, wheezing, tachycardia, hypotension and skin rashes of urticarial non-specific maculopapular type can occur within a few minutes of administration. Serious reactions seem to be controlled by IV adrenaline, corticosteroids or antihistamines.

Serum Electrolyte Disturbances: Hypomagnesaemia, hypocalcaemia, hyponatraemia, hypokalaemia and hypophosphataemia have been reported to occur in patients treated with cisplatin and are probably related to renal tubular damage. Hypomagnesaemia and hypocalcaemia may result in tetany. Generally, normal serum electrolyte levels are restored by administering supplemental electrolytes and discontinuing cisplatin. Inappropriate antidiuretic hormone syndrome has also been reported.

Neurotoxicity: Usually characterised by peripheral neuropathies and paresthesia in both upper and lower extremities. Peripheral neuropathy, while reversible, may take a year or more to recover. Losses of taste and seizures have also been reported. Neuropathies resulting from cisplatin treatment may occur after prolonged therapy; however, neurological symptoms have been reported to occur after a single dose. The neuropathy may progress after stopping treatment.

Hyperuricaemia: Hyperuricaemia occurring with cisplatin is more pronounced with doses greater than 50 mg/m². Allopurinol effectively reduces uric acid levels.

Other Toxicities: Vascular toxicities coincident with the use of cisplatin in combination with other antineoplastic agents have been reported rarely. These events may include myocardial infarction, cerebrovascular accident, thrombotic microangiopathy (haemolytic uraemic syndrome) or cerebral arteritis. There are also reports of Raynaud's phenomenon occurring in patients treated with the combination of bleomycin, vinblastine with or without cisplatin. It has been suggested that hypomagnesaemia developing coincident with the use of cisplatin may be an added, although not essential factor, associated with this event. However the cause of this Raynaud's phenomenon is currently unknown. Other toxicities reported to occur infrequently are cardiac abnormalities including tachycardia and arrhythmia.

Local soft tissue toxicity has been reported rarely following extravasation of cisplatin. Infiltration of solutions of cisplatin may result in tissue cellulitis, fibrosis and necrosis.

OVERDOSE AND TREATMENT

Overdosage can be expected to cause the toxic effects described above, but to an exaggerated degree. Adequate hydration and osmotic diuresis may help reduce the toxicity of cisplatin if administered promptly following overdosage.

Convulsions may be treated with appropriate anticonvulsants.

Renal function, cardiovascular function and blood counts should be monitored daily in order to assess the potential toxicity to these systems.

Serum magnesium and calcium levels should be carefully monitored as should symptoms and signs of voluntary muscle irritability.

If symptomatic tetany develops, electrolyte supplements should be administered.

Serum liver enzymes and uric acid should also be monitored daily after an acute overdose.

If fever develops during prolonged myelosuppression, appropriate presumptive antibiotic coverage should be instilled after cultures have been obtained.

STORAGE CONDITION

Store in a cool place (30 °C). Protect from light. Do not refrigerate.

INCOMPATIBILITIES

Cisplatin may interact with aluminium to form a black precipitate. Needles, syringes, catheters or IV administration sets that contain aluminium parts which may come in contact with cisplatin should not be used for preparation or administration of the drug. The stability of cisplatin is adversely affected by the presence of bisulphite, metabisulphite, sodium bicarbonate and β uorouracil.

SHELF LIFE

3 years

After dilution: 24 hours at 20-25°C.

SPECIAL PRECAUTIONS FOR DISPOSAL AND OTHER HANDLING

Single use only. Discard any unused contents.

Refer to local cytotoxic handling guidelines.

Dilution:

Cisplatin 1 mg/ml Sterile Concentrate should be diluted in 2 litres of 0.9% sodium chloride injection.

Administration:

Should be administered only by or under the direct supervision of a qualified physician who is experienced in the use of cancer chemotherapeutic agents.

Preparation (Guidelines):

1. Chemotherapeutic agents should be prepared for administration only by professionals who have been trained in the safe use of the preparation.
2. Operations such as reconstitution, dilution and transfer to syringes should be carried out only in the designated area.
3. The personnel carrying out these procedures should be adequately protected with clothing, gloves and eye shield.
4. Pregnant personnel are advised not to handle chemotherapeutic agents.

Contamination:

- (a) In the event of contact with the skin or eyes, the affected area should be washed with copious amounts of water or normal saline. A bland cream may be used to treat the transient stinging of skin. Medical advice should be sought if the eyes are affected.
- (b) In the event of spillage, operators should put on gloves and mop up the spilled material with a sponge kept in the area for that purpose. Rinse the area twice with water. Put all solutions and sponges into a plastic bag and seal it.

Disposal:

Syringes, container, absorbent materials, solution and any other contaminated material should be placed in a thick plastic bag or other impervious container and incinerated.

DOSAGE FORMS AND PACKAGING AVAILABLE

Cisplatin Injection 1mg/ml is available as 10 ml and 50 ml single dose amber vials, individually packed in a carton.

Manufactured by:



INTAS PHARMACEUTICALS LTD.

Plot No: 457-458, Sarkhej- Bavla Highway,
Matoda, Sanand, Dist- Ahmedabad,
Gujarat, India- 382 210.

Product Registration Holder :

Jetpharma Sdn Bhd

No.13, Jalan Rajawali 2,
Bandar Puchong Jaya
47100 Puchong
Selangor, Malaysia

Date of revision: April 2024

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